

Choosing the knobs

Tuning, folds, and what cross-validation is actually for

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Where we are

Lecture 3, on k in k -NN:

*" k is what is called a **tuning parameter**; in this case it governs how much to smooth our function."*

Lecture 4 closed on two tasks:

Model selection — *choosing a model's shape or its tuning parameters.*

Model assessment — *how good is the model we chose?*

Today: the first task on real data, and what the second one costs when the folds are drawn carelessly.

Parameters and hyperparameters

Parameter. $\hat{\beta} = \arg \min_b \text{RSS}(b)$. The data pick it.

Hyperparameter. k . Polynomial degree. K , the number of folds. Tomorrow, λ .

Why can the data not pick these the same way?

Parameters and hyperparameters

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Why can the data not pick these the same way?

Because training MSE is **monotone in flexibility**. Minimise it over k and you get $k = 1$, every time — lecture 3's high-variance corner. Over the degree, the highest degree on offer.

So the choice is made on rows the model has not seen. That is the second loop.

The knobs you already own

knob	where you met it	what it controls
k	lecture 3	how much to smooth
degree	lecture 4	how curved f may be
K	lecture 4	bias–variance of the CV estimate itself
λ	tomorrow	how hard to shrink
the folds	today	what “out of sample” means

The last one is not usually called a hyperparameter. It behaves exactly like one.

Part 1 · The knob you set without noticing

The most common model in political science

You have a treatment D , a moderator X , and a theory that the effect of D depends on X . So you fit

$$Y = \beta_0 + \beta_D D + \beta_X X + \beta_{DX} (D \times X) + \varepsilon$$

and report the marginal effect

$$\widehat{ME}(x) = \hat{\beta}_D + \hat{\beta}_{DX} x$$

That is **a straight line in x** , imposed before you looked at anything.

Nobody calls it a tuning parameter. It is one: flexibility, fixed at its most rigid setting, by convention.

Hainmueller, Mummolo & Xu (2019)

Political Analysis 27(2), “How Much Should We Trust Estimates from Multiplicative Interaction Models?”

Their argument, in our vocabulary: the linear interaction model is an untuned choice, and it fails in two ways.

1. The true $ME(x)$ may not be linear in x .
2. There may be almost no data in parts of the X range, so the line is **extrapolating**, not estimating.

Their remedy is one you already know: **let the flexibility be chosen**. Bin the moderator, or use a kernel estimator whose bandwidth is picked **by cross-validation**.

Vernby (2013), *AJPS*

Sweden extends the local franchise to noncitizens. Does school spending follow?

- 183 municipalities
- D = noncitizen share of the electorate
- X = tax base, the moderator
- Y = change in education spending

```
u <- paste0("https://raw.githubusercontent.com/xuyiqing/",  
            "interflex/master/data/interflex.RData")  
if (!file.exists("interflex.RData")) download.file(u, "interflex.RData", mode="wb")  
load("interflex.RData")  
v <- app_vernby2013  
dim(v)
```

```
[1] 183    9
```

The published shape

```
m <- lm(school_diff ~ noncitvotsh * Taxbase2, data = v)
round(summary(m)$coefficients, 3)
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.256	0.456	0.563	0.574
noncitvotsh	39.938	10.499	3.804	0.000
Taxbase2	0.437	0.526	0.831	0.407
noncitvotsh:Taxbase2	-37.427	11.843	-3.160	0.002

Table 2: Implied marginal effect across the tax base

p05	p25	p50	p75	p95
11.7	10.5	9.5	6.6	-2.7

A clean story: enfranchisement raises school spending in poor municipalities and **lowers** it in rich ones — the interaction carries $t = -3.16$.

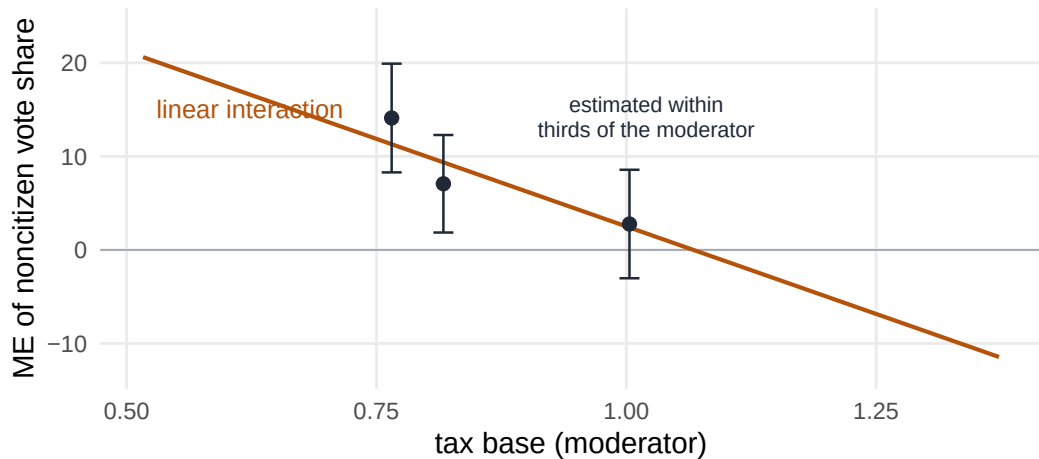
Now relax the knob

HMX's binning estimator: cut the moderator into thirds, estimate the effect separately inside each. No functional form across bins.

```
v$bin <- cut(v$Taxbase2, quantile(v$Taxbase2, c(0, 1/3, 2/3, 1)),
           labels = c("low", "mid", "high"), include.lowest = TRUE)
binest <- do.call(rbind, lapply(levels(v$bin), function(L) {
  z <- v[v$bin == L, ]; f <- lm(school_diff ~ noncitvotsh, data = z)
  data.frame(third = L, n = nrow(z), xbar = mean(z$Taxbase2),
            ME = coef(f)[2], se = summary(f)$coefficients[2, 2])
}))
knitr::kable(binest, digits = 2, row.names = FALSE)
```

third	n	xbar	ME	se
low	61	0.77	14.10	2.96
mid	61	0.82	7.07	2.66
high	61	1.00	2.77	2.96

Now relax the knob



What actually happened

The line has to keep going. The data stop.

- $\widehat{ME}$ at the top of the range: -2.7 from the line, $+2.8 \pm 5.8$ from the data in that region.
- The negative number is not an estimate. It is the straight line continuing past where anything was observed.
- The $t = -3.16$ on the interaction is real, and it is about the **slope** — not about any municipality crossing zero.

The rigid setting did not just lose precision. It produced a sign that is not in the data.

The other outcome: Huddy, Mason & Aarøe (2015), *APSR*

“Expressive Partisanship.” Does partisan identity amplify anger at threats to one’s party?

```
h <- app_hma2015
mh <- lm(totangry ~ threat * pidentity, data = h)
round(summary(mh)$coefficients, 3)
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.278	0.040	7.025	0.000
threat	0.018	0.055	0.328	0.743
pidentity	-0.032	0.055	-0.581	0.561
threat:pidentity	0.421	0.076	5.506	0.000

$n = 1482$, threat randomised, $\hat{\beta}_{DX} = 0.421$ (0.076), $t = 5.51$.

Same check, with lecture 3’s tool: estimate $E[Y | X]$ **separately in each arm** by k -NN, choose k by 10-fold CV. The marginal effect is the gap between the two local averages.

Choosing k inside each arm

```
knn1 <- function(xtr, ytr, xte, ks) {                                # all k at once
  M <- outer(xte, xtr, function(a, b) (a - b)^2)
  out <- matrix(NA_real_, length(xte), length(ks))
  for (i in seq_along(xte)) {
    o <- order(M[i, ]); out[i, ] <- (cumsum(ytr[o]) / seq_along(o))[ks]
  }
  out
}

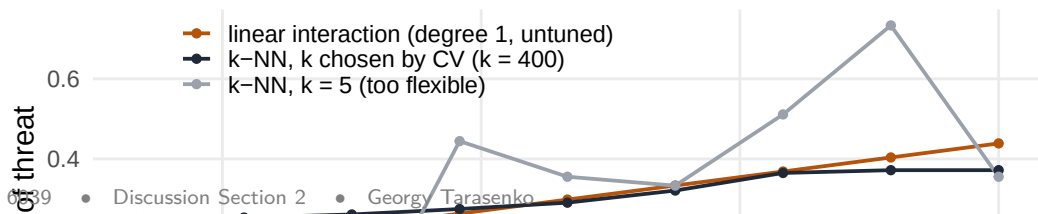
cvk1 <- function(x, y, ks, K = 10) {                                # lecture 4, verbatim
  set.seed(6039); f <- sample(rep(1:K, length.out = length(y)))
  M <- matrix(NA_real_, K, length(ks))
  for (j in 1:K) M[j, ] <- colMeans((y[f == j] -
                                     knn1(x[f != j], y[f != j], x[f == j], ks))^2)
  colMeans(M)
}

ks <- c(5, 10, 25, 50, 100, 200, 400)
best <- sapply(0:1, function(d) {
```

One dataset, three answers

```
g <- seq(quantile(h$pidentity, .05), quantile(h$pidentity, .95), length.out = 10)
me <- function(k) knn1(h$pidentity[h$threat == 1], h$totangry[h$threat == 1], k) -
  knn1(h$pidentity[h$threat == 0], h$totangry[h$threat == 0], k)
rbind(`linear interaction` = bh["threat"] + bh["threat:pidentity"] * g,
      `k-NN, CV's k`      = me(max(best)),
      `k-NN, k = 5`      = me(5)) |> round(2)
```

	[,1]	[,2]	[,3]	[,4]	[,5]	[,6]	[,7]	[,8]	[,9]
linear interaction	0.16	0.19	0.23	0.26	0.30	0.33	0.37	0.40	0.44
k-NN, CV's k	0.25	0.25	0.26	0.28	0.29	0.32	0.37	0.37	0.37
k-NN, k = 5	0.11	0.04	-0.07	0.44	0.36	0.33	0.51	0.73	0.36



One dataset, three answers

setting	what you would report	
$k = 5$, untuned	effect non-monotonic, turns negative	noise
degree 1, untuned	effect rises steeply, 0.16 → 0.44	overstated slope
k from CV	effect rises gently, 0.25 → 0.37	what the data support

Two lessons, and the second is the one people forget:

1. An untuned knob can manufacture a finding — in either direction.
2. Here, tuning largely **vindicates** the published paper. That is a result. You only get to say it because you checked.

Part 1, in one line

Tuning is not “make the model fancier.”

It is letting the data set the flexibility — and being willing to hear the answer “stay simple.”

HMX ran this check on **46 interaction effects from 22 articles** in the *APSR*, *AJPS*, *CPS*, *IO* and *JOP*, published 2006–2015.

	share of cases
the effect changes linearly in the moderator	48%
cannot reject that the effect is equal at typical low and high moderator values	70%

And the knob is still being argued about

This is not settled methodology from a textbook. It is a live fight.

- **Simonsohn (2024)**, *AMPPS*, and the companion “Dear Political Scientists: Don’t Bin, GAM Instead”: binning is the wrong fix; model the curve with a GAM.
- **Hainmueller, Liu, Liu, Mummolo & Xu (2025)**: the critique benchmarks against the wrong estimand; the kernel estimator does recover the conditional marginal effect; GAM is for exploration, not for this.

Both sides agree on what matters today: **the flexibility is a choice, it has to be made explicitly, and the data have to be allowed to inform it.**

They disagree about *how*. That disagreement is open, and it is open to you.

Part 2 · The folds are a knob too

A question you can actually ask

How much of a country's level of democracy is accounted for by its structural conditions?

```
uv <- paste0("https://raw.githubusercontent.com/vdeminstitute/",
             "vdemdata/master/data/vdem.RData")
if (!file.exists("vdem_slim.rds")) {
  if (!file.exists("vdem.RData")) download.file(uv, "vdem.RData", mode = "w")
  load("vdem.RData")
  saveRDS(vdem[, c("country_name", "year", "v2x_polyarchy", "e_gdppc", "e_pop",
                  "e_total_resources_income_pc", "e_regionpol_6C")], "vdem_slim.rds")
  rm(vdem); gc()
}
vd <- readRDS("vdem_slim.rds")
vd <- vd[vd$year >= 1960 & vd$year <= 2023, ]; vd <- vd[complete.cases(vd), ]
vd$lgdp <- log(pmax(vd$e_gdppc, .01)); vd$lpop <- log(pmax(vd$e_pop, .01))
vd$lres <- log1p(pmax(vd$e_total_resources_income_pc, 0))
regmat <- model.matrix(~ factor(e_regionpol_6C) - 1, data = vd)[, -1, drop = FALSE]
X <- cbind(as.matrix(vd[, c("lgdp", "lpop", "lres", "year")]), regmat)
```

k-NN and 10-fold CV, exactly as on Tuesday

```
knn_all <- function(Xtr, ytr, Xte, ks) {  
  mu <- colMeans(Xtr); sdv <- apply(Xtr, 2, sd); sdv[sdv == 0] <- 1  
  A <- scale(Xtr, mu, sdv); B <- scale(Xte, mu, sdv) # standardise INSIDE  
  Dm <- outer(rowSums(B^2), rowSums(A^2), "+") - 2 * B %*% t(A)  
  out <- matrix(NA_real_, nrow(B), length(ks))  
  for (i in seq_len(nrow(B))) {  
    o <- order(Dm[i, ]); out[i, ] <- (cumsum(ytr[o]) / seq_along(o))[ks]  
  }  
  out  
}  
  
cv <- function(fold, ks) {  
  M <- matrix(NA_real_, max(fold), length(ks))  
  for (j in 1:max(fold)) {  
    tr <- fold != j; te <- fold == j  
    M[j, ] <- colMeans((y[te] - knn_all(X[tr, ], y[tr], X[te, ], ks))^2)  
  }  
  colMeans(M)
```

Modernization theory, solved

k	CV MSE	implied R^2
1	0.0026	0.968
2	0.0016	0.980
3	0.0023	0.972
5	0.0032	0.960
10	0.0068	0.916
25	0.0139	0.828
50	0.0197	0.756
100	0.0246	0.697
250	0.0306	0.622
500	0.0346	0.573

$k = 2$ wins, at MSE 0.0016 — an implied R^2 of **0.98**.

Five structural variables and a local average of two neighbours account for 98% of the variation in democracy across the postwar world.

Do we believe this?

Five variables. Two neighbours. $R^2 = 0.98$.

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The folds were drawn at random over **country-years**.

Denmark 1994 and Denmark 1995 are nearly the same row. A random fold puts one in training and the other in the held-out set.

Do we believe this?

Five variables. Two neighbours. $R^2 = 0.98$.

The folds were drawn at random over **country-years**.

Denmark 1994 and Denmark 1995 are nearly the same row. A random fold puts one in training and the other in the held-out set.

So hold out whole **countries** instead.

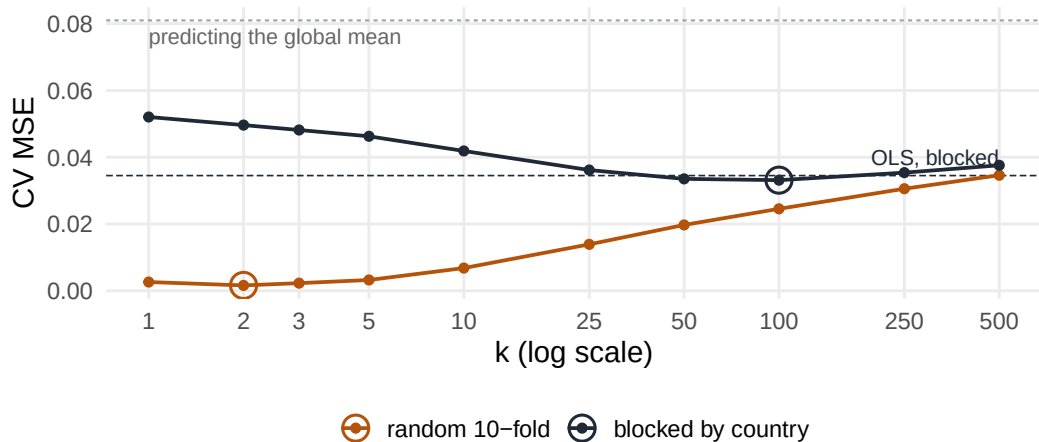
Draw the folds by country instead

	CV MSE	implied R2
scored on held-out country-years	0.0016	0.980
scored on held-out countries	0.0496	0.387

The same $k = 2$ model. **31 times worse.**

Nothing changed but which rows went into which fold.

The whole picture



Circles mark the k each procedure selects.

Two things moved, not one

	random folds	blocked by country
the k that wins	2	100
its CV MSE	0.0016	0.0331
OLS	0.0327	0.0345

1. The **error estimate** was inflated by a factor of 31.
2. The **hyperparameter** the procedure selected moved from $k = 2$ to $k = 100$.
3. Under honest folds, k -NN at its best (0.0331) barely edges plain OLS (0.0345).
The flexible model buys almost nothing.

If you tune with leaky folds you do not merely overstate your error. **You choose the wrong model.**

Why $k = 2$ wins on random folds

The two nearest neighbours of Denmark 1995 are Denmark 1994 and Denmark 1996.

$k = 2$ is not a model of democratisation. It is a **lookup table keyed on country identity**, and random folds hand it the key.

Block by country and the key is gone, so the estimator has to fall back on structure — which is worth an R^2 of about 0.59, not 0.98.

The rule, restated

Tuesday's version:

The fold must redo every data-dependent step: screening, standardising, imputing, choosing k .

Today's addition:

And the folds must respect the dependence in your data.

Random K -fold assumes the rows are exchangeable draws. Ask whether yours are before you accept the number.

Where this bites in your own work

your data	the unit to hold out
country-year or state-year panel	country / state
survey with respondents seen twice	respondent
students in schools, voters in precincts	school / precinct
districts in space	a spatial block
dyads	both members of the dyad
anything with a time order and a forecast	the future

If two rows would never be separated in the real prediction problem, they should not be separated by your folds.

What to take away

Selection and assessment, one more time

The number that chose your model cannot also be the number that grades it.

- Selection: an inner loop, over the grid, on training rows only.
- Assessment: an outer loop, or a test set you touch once.
- Do both with one set of folds and the reported error is the best of many draws.

This is Tuesday's slide 19 and today's Part 2. Same disease, two organs.

What to report

1. The **grid** you searched, and how many fits it took.
2. The **metric**, and why it suits the outcome.
3. **How the folds were drawn**, and what unit was held out.
4. Whether the reported error came from the **same loop** that chose the model.
5. For any interaction: the **support** across the moderator, and one estimate that does not assume linearity.

None of this is expensive. All of it is usually missing.

Three sentences you could have written today

1. *“Noncitizen enfranchisement reduces school spending in wealthy municipalities.”* — an artifact of the straight line, at a tax base where no municipality was observed.
2. *“Structural conditions account for 98% of the variation in democracy.”* — an artifact of the folds, which handed the model the country’s own past.
3. *“The effect declines in the tax base, and flexible estimators do not beat OLS on V-Dem.”* — what survives.

The first two are more interesting. That is the problem.

For tomorrow and for the problem set

- Tomorrow is penalisation: λ is the same object as k , and is chosen the same way. Everything today transfers.
- On PS1: when you fit anything with a knob, show the grid and say how the folds were drawn.
- If your data have repeated units, block on them and report both numbers. The gap is itself a finding.

Questions?

References I

- Hainmueller, J., J. Mummolo & Y. Xu. 2019. "How Much Should We Trust Estimates from Multiplicative Interaction Models? Simple Tools to Improve Empirical Practice." *Political Analysis* 27(2): 163–192.
- Vernby, K. 2013. "Inclusion and Public Policy: Evidence from Sweden's Introduction of Noncitizen Suffrage." *AJPS* 57(1): 15–29.
- Huddy, L., L. Mason & L. Aarøe. 2015. "Expressive Partisanship: Campaign Involvement, Political Emotion, and Partisan Identity." *APSR* 109(1): 1–17.
- Simonsohn, U. 2024. "Interacting with Curves: How to Validly Test and Probe Interactions in the Real (Nonlinear) World." *AMPPS* 7(1).
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- Hastie, T., R. Tibshirani & J. Friedman. 2009. *The Elements of Statistical Learning*, 2nd ed., §7.10.2.

References II

Roberts, D. R. et al. 2017. "Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure." *Ecography* 40(8): 913–929.

Application data via the `interflex` package (Hainmueller, Mummolo, Xu & Liu).